DevOps Linux Fundamentals (Day 7) - Machine Test

Real-World Scenario: You have joined as a Junior DevOps Engineer. A production Ubuntu server requires package management, storage verification, mount management, backup operations, and troubleshooting. Complete all tasks and submit command outputs/screenshots where required.

Part 1 - Package Management

1. Update package repository information.
sudo apt update

2. Install tree, curl, and git packages.

3. Search for Docker-related packages.
apt search docker

4. Display package information for nginx.
apt show nginx

5. Remove tree package.
sudo apt remove tree

Evidence: Capture outputs of apt search docker and apt show nginx.

Part 2 - Storage Analysis

1. Check filesystem usage using df -h.
2. Check project folder size using du -sh project.
3. View storage devices using lsblk.
4. Display UUID information using sudo blkid.
5. Display partition details using sudo fdisk -l.

Save outputs into storage-report.txt.

Part 3 - Mounting & Filesystem Management

1. Create a mount point: sudo mkdir /mnt/devops-data
2. Display current mount information using mount.
3. Check mounted filesystems using df -h.
4. View automatic mount configuration using cat /etc/fstab.
5. Create mount-report.txt containing mount, df -h, and /etc/fstab output.

Part 4 - Backup & Compression

1. Create TAR backup.
tar -cvf project-backup.tar project/

2. Extract TAR backup.
tar -xvf project-backup.tar

3. Create compressed archive.
tar -czvf project-backup.tar.gz project/

4. Extract compressed archive.
tar -xzvf project-backup.tar.gz

5. Create ZIP backup.
zip -r project.zip project/

6. Extract ZIP backup.
unzip project.zip

7. Backup application using rsync.
rsync -av project/ backup/

Part 5 - Production Troubleshooting Challenge

Scenario:
Filesystem Size Used Avail Use%
/dev/sda1 50G 49G 1G 98%

Answer:
1. Which command identifies disk usage?
2. Which command shows folder size?
3. Which command shows available disks?
4. Which command removes unused packages?
5. Which command is commonly used for production backups?

Bonus Challenge

Create a backup workflow:
mkdir deployment-backup
rsync -av project/ deployment-backup/
tar -czvf deployment-backup.tar.gz deployment-backup/

Verify using:
ls -lh deployment-backup.tar.gz

Interview Questions

1. What is APT?
2. Difference between apt update and apt upgrade?
3. What does df -h display?
4. Difference between df -h and lsblk?
5. What is UUID?
6. What is mounting in Linux?
7. Difference between mount and umount?
8. What is /etc/fstab?
9. Difference between .tar and .tar.gz?
10. Why is rsync preferred in production?